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Module: Internet of Things and Embedded System Practice

Smart Room Temperature Monitoring System - IoT Assignment

Step 1: IoT Topic

Smart environment monitoring for a classroom or room using temperature and humidity sensing.

Step 2: Identify Objects

General Object:

Classroom environment monitoring system.

Specific Objects:

- DHT11 temperature and humidity sensor
- ESP32 microcontroller
- WiFi network
- Cloud dashboard
- Power supply

Step 3: IoT Components

Sensors:

DHT11 temperature and humidity sensor.

Microcontroller:

ESP32.

Connectivity Module:

Built-in WiFi in ESP32.

Actuator:

LED for temperature alert.

Power Supply:

5V USB power.

Step 4: Power Consumption Calculation

DHT11 sensor:

Current = 0.3 mA at 5V

Power = $V \times I = 5 \times 0.0003 = 0.0015 \text{ W}$

ESP32 with WiFi active:
Current = 240 mA at 5V
Power = $5 \times 0.24 = 1.2 \text{ W}$

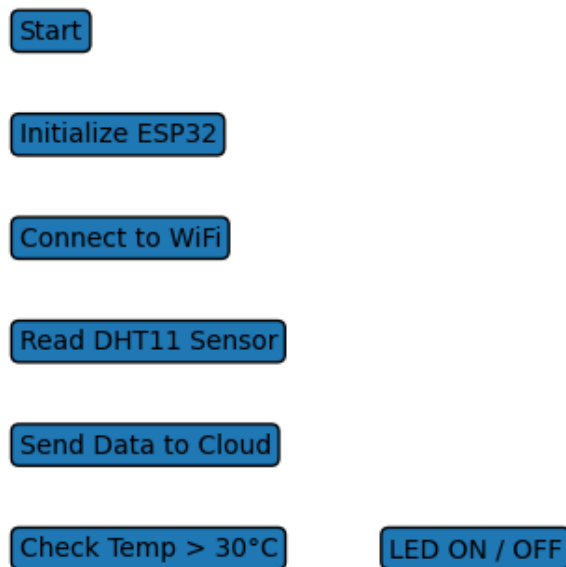
LED:
Current = 20 mA at 5V
Power = $5 \times 0.02 = 0.1 \text{ W}$

Total Power Consumption:
 $0.0015 + 1.2 + 0.1 = 1.3015 \text{ W}$

Step 5: System Flowchart Description

Start → Initialize ESP32 → Connect to WiFi → Read temperature and humidity from DHT11 → Send data to cloud dashboard → Check temperature threshold → If temperature > 30°C turn ON LED → Else turn OFF LED → Wait 10 seconds → Repeat

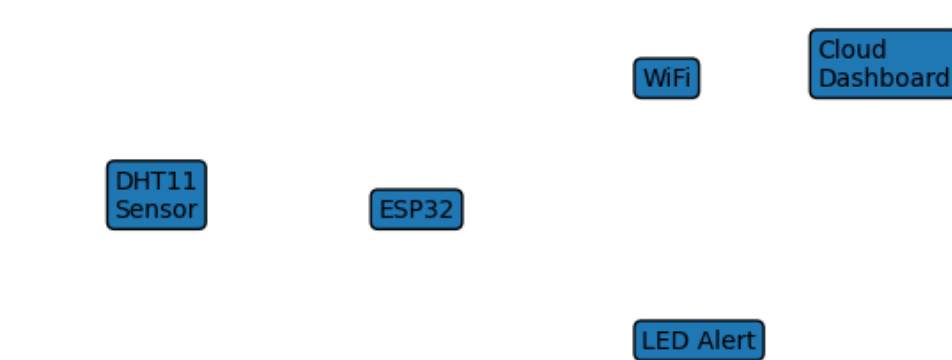
Flowchart Diagram



Block Diagram Description

DHT11 Sensor → ESP32 Microcontroller → WiFi → Cloud Dashboard
ESP32 → LED Alert

Block Diagram



Sample ESP32 Code (for demonstration)

```
#include <WiFi.h>
#include "DHT.h"

#define DHTPIN 4
#define DHTTYPE DHT11
#define LEDPIN 2

DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(115200);
  dht.begin();
  pinMode(LEDPIN, OUTPUT);
}

void loop() {
  float temp = dht.readTemperature();
  if (temp > 30) {
    digitalWrite(LEDPIN, HIGH);
  } else {
    digitalWrite(LEDPIN, LOW);
  }
}
```

```
    delay(10000);  
}
```

Step 6: Conclusion

This system monitors room temperature and humidity in real time. Data is sent to a cloud dashboard for remote viewing. An LED provides a local alert when temperature exceeds the threshold.

Benefits:

- Low cost implementation
- Real time monitoring
- Improves classroom comfort

Challenges:

- Dependence on WiFi connectivity
- Limited accuracy of DHT11 sensor
- Power consumption during WiFi transmission

Future Improvements:

- Add LCD display
- Store historical data in a database
- Send mobile notifications
- Add automatic fan control